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PRODUCT DATASHEET · POWER SYSTEM TRAINING SIMULATOR

# Karuma HPP Digital Twin

Generation – Transmission – Distribution simulator with integrated SCADA, protection relays, 3D training lab and instructor station




Real-time digital twin · Newton-Raphson load flow · Electromechanical dynamics · 65 protection functions · SCADA HMI · 3D training lab · Instructor & trainee workstations · Windows 10/11 · .exe

ASFAN

**Software designed & developed by ASFAN**

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Software version 1.0.0

## Product overview

The **Karuma HPP Digital Twin** is a complete, real-time software replica of the Karuma hydro power plant training system. It reproduces the entire power chain — **prime mover, synchronous generator, step-up transformer, 220 kV transmission lines, substation switchgear & protection, distribution transformers with voltage regulation and distribution loads** — and presents it through a professional SCADA interface, protection relay panels, an interactive 3D training lab and a dedicated instructor station.

Trainees **monitor V, I, MW, MVAr and Hz**, **control breakers, loads and generation**, **analyse power flows** and **run training scenarios and fault conditions** exactly as they would on the physical training laboratory — safely, repeatably and without equipment wear. Every quantity is calculated by a physics-based network and machine model, not by animations or pre-recorded data.

| FREQUENCY | TOTAL GENERATION | TOTAL LOAD | VOLTAGE (HV) |
|---|---|---|---|
| 50.00 Hz | 120 MW | 118.5 MW | 220 kV |

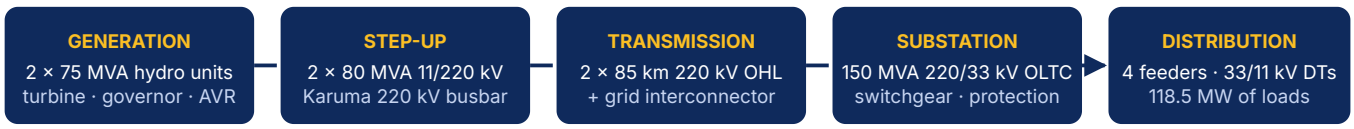
### Key features

- Physical power system training simulator: turbine trainer, generator, step-up transformer, transmission line simulator, substation, distribution transformer and loads
- SCADA system: Overview, Generation, Transmission, Distribution, Alarms, Reports and Trends pages
- Generator Control & AVR, Protection Relays, AGC & Network Control and SCADA interface panels
- Training room set-up: one instructor workstation and multiple trainee workstations over the LAN; in the 3D lab each of the 14 desk monitors shows its own live page
- Real-time electromechanical simulation with automatic voltage, frequency and tap-changer control
- Short-circuit faults of any type at any location, cleared by realistic numerical protection

### Typical applications

- Operator training for hydro power plants, transmission and distribution utilities
- University and technical-college power system laboratories
- Protection engineering courses (distance, differential, overcurrent, UFLS, auto-reclose)
- Black-start, islanding and grid-synchronisation drills
- Pre-commissioning familiarisation with the SCADA and switching procedures
- Trainee assessment with automatic scoring and printable reports

### System architecture




**SIMULATION CORE**
Node-breaker topology processor · Newton-Raphson load flow per island · machine, turbine & control dynamics
Symmetrical-component fault analysis · protection relay engine · SCADA historian · alarms & SOE
Scenario assessment engine · classroom network server (instructor ⇄ trainee workstations)

## SCADA – Power system overview

The SCADA overview shows the complete single-line diagram from the generator terminals to the customer loads, with live, colour-coded voltage levels, animated power flow, breaker states and the four key system indicators requested by the operator: **Frequency, Total Generation, Total Load and Voltage (HV)**. Any breaker can be operated directly from the diagram (with interlocking and synchro-check); clicking an equipment item opens its detailed page.

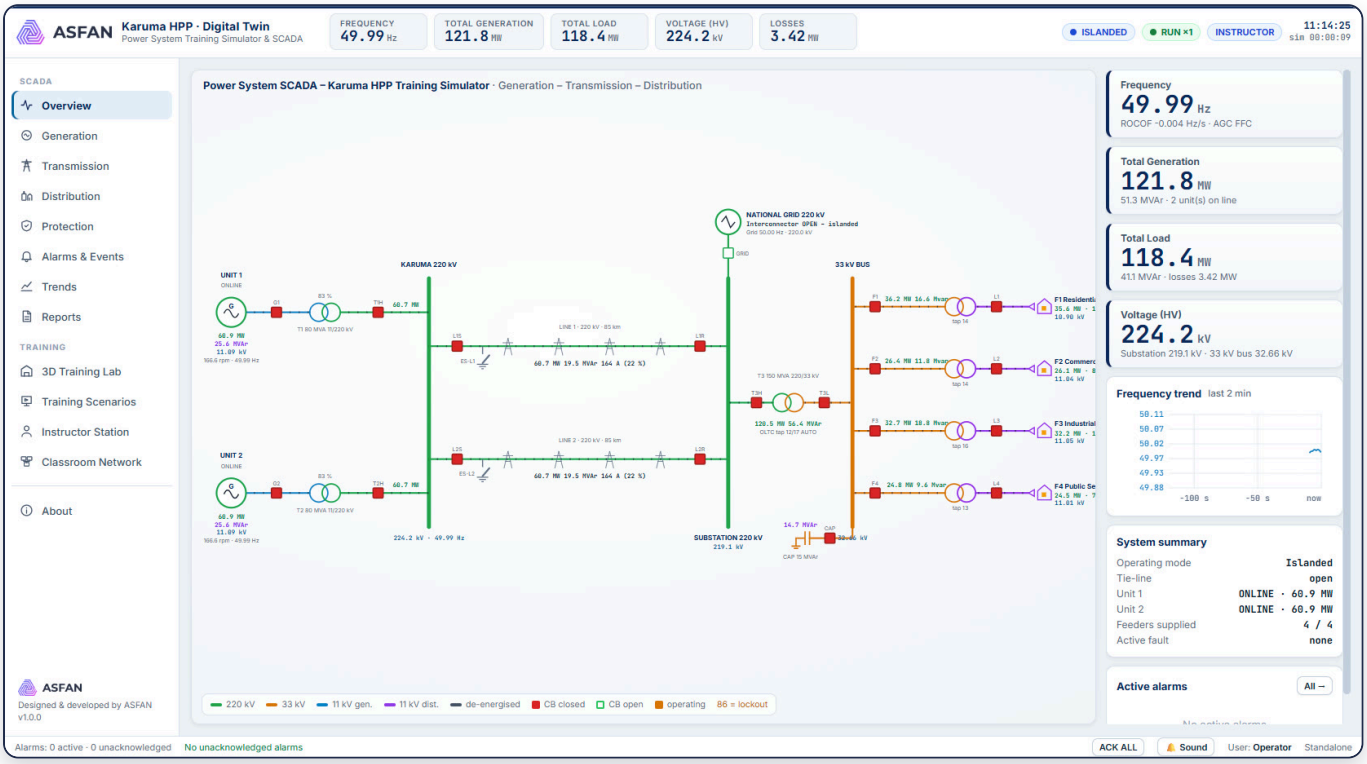



*Figure 1 – SCADA overview: single-line diagram, key indicators, frequency trend, system summary and active alarms.*

### Simulated network

| Section | Equipment represented |
|---|---|
| Generation | 2 × 75 MVA, 11 kV, 0.85 pf hydro generators (63.75 MW each), Francis turbines with penstock, governors, static excitation & AVR, generator circuit breakers |
| Step-up | 2 × 80 MVA, 11/231 kV Dyn11 step-up transformers with off-circuit taps, 220 kV breakers, Karuma 220 kV busbar |
| Transmission | 220 kV double-circuit overhead line, 2 × 85 km, line breakers at both ends, line earthing switches, 2 500 MVA national-grid interconnector |
| Substation | Substation 220 kV busbar, 150 MVA 220/33 kV YNyn0 power transformer with on-load tap changer (±8 × 1.25 %), 33 kV busbar, 15 MVAr capacitor bank |
| Distribution | 4 × 33 kV feeders (4–6 km), 4 distribution transformers 31.5–50 MVA 33/11 kV with OLTC & automatic voltage control, 11 kV incomers |
| Loads | Residential (36 MW), Commercial (26 MW), Industrial with motor load (32 MW), Public services & hospital (24.5 MW) – voltage- and frequency-dependent |

## Generation – Generator Control & AVR

Each unit has a full control faceplate reproducing the **Generator Control & AVR panel**: start / stop / emergency-stop sequences, speed and load set-points, governor mode (droop or isochronous), field breaker and AVR (auto / manual), synchroscope with manual and automatic synchronising, P–Q capability diagram and all turbine and generator measurements.

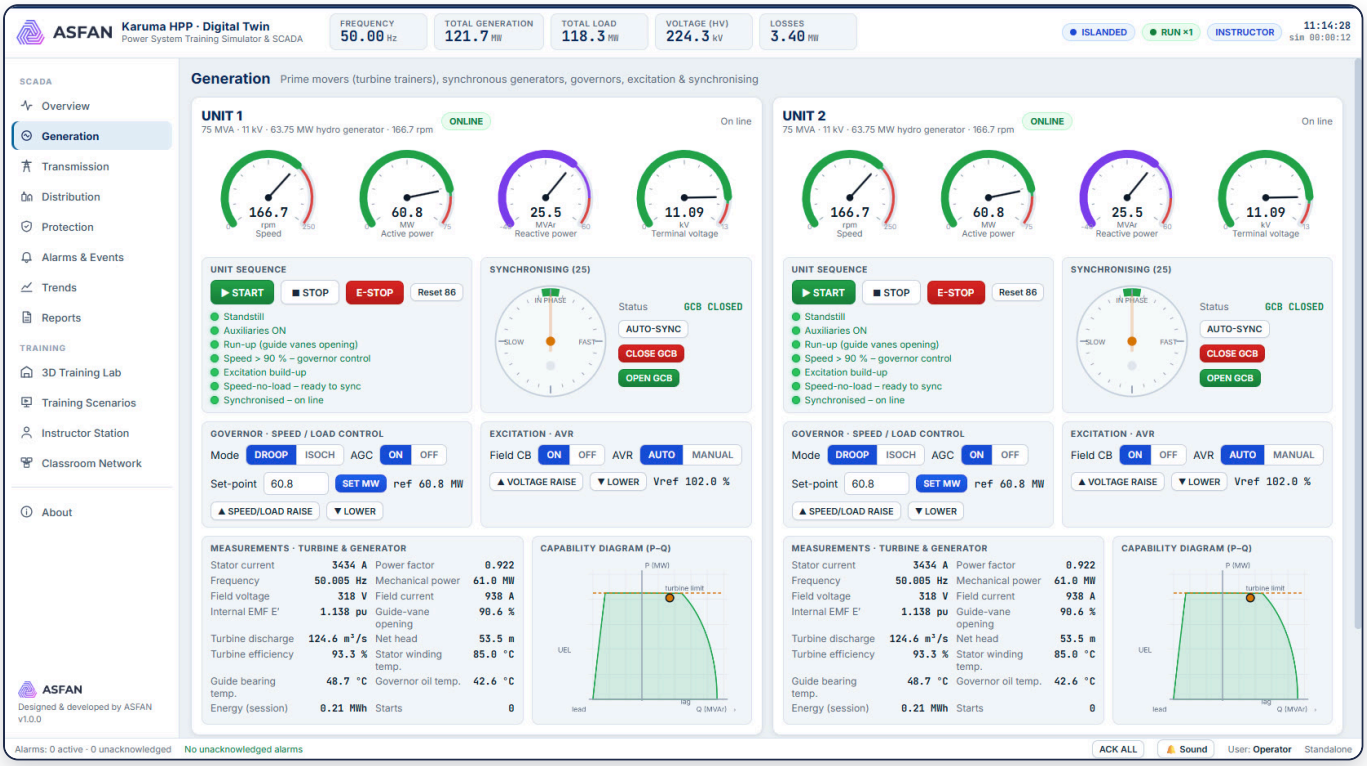



*Figure 2 – Generation page with two unit faceplates, gauges, start sequence, synchroscope and capability diagram.*

| Synchronous generator & turbine (per unit) | |
|---|---|
| Rating | 75 MVA, 11 kV, pf 0.85, 63.75 MW |
| Speed | 166.7 rpm, 36 poles, 50 Hz |
| Reactances | Xd 1.05 · Xd' 0.30 · Xd'' 0.22 · X2 0.22 pu |
| Time constants | T'd0 6 s · H 4.0 s |
| Prime mover | Francis turbine, 54 m net head, water-hammer (Tw) model, guide-vane servo with rate limits |
| Measurements | V, I, MW, MVAr, pf, Hz, rpm, gate %, discharge, head, efficiency, field V/I, temperatures, energy |

| Controls | |
|---|---|
| Governor | PI speed/load controller, 4 % droop or isochronous, load ramp limiter, load-reference feed-forward |
| Excitation / AVR | Static exciter, PI AVR with reactive droop, V/Hz limiter, manual field control, field breaker |
| Synchronising | Synchroscope (Δθ, Δf, ΔV), synchro-check relay 25, auto-synchroniser |
| AGC & network control | OFF / flat-frequency / tie-line-bias modes, frequency bias, interchange schedule, ACE display |
| Sequences | Auxiliaries → run-up → speed control → excitation → speed-no-load → synchronised |

# Transmission & distribution




*Figure 3 – Transmission: step-up transformers, 220 kV lines, grid synchronising, voltage profile, N-1 and short-circuit studies.*

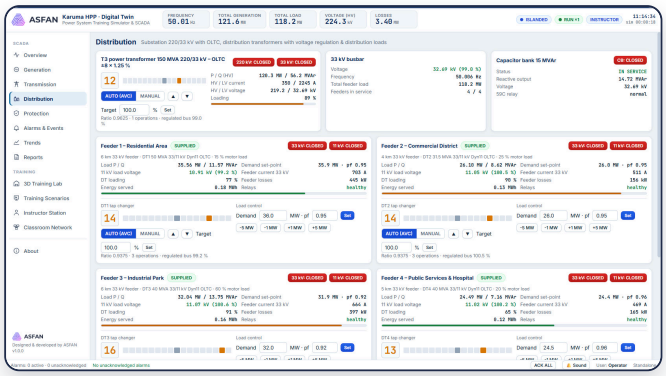



*Figure 4 – Distribution: T3 OLTC, capacitor bank, four feeders with tap changers and load control.*

## Transmission functions

- Sending / receiving P, Q, current, losses and loading of each line and transformer
- Line parameters: Z1 = 0.06 + j0.41 Ω/km, Z0 = 0.30 + j1.15 Ω/km, shunt charging; rating 300 MVA
- Grid interconnector with synchroscope, synchro-check and grid frequency/voltage set by the instructor
- Busbar voltage magnitudes and angles; voltage profile from the generator to the load
- **N-1 contingency analysis** and **short-circuit level study** (3-phase and 1-phase, S″k, X/R) on demand, exportable to CSV

## Distribution functions

- On-load tap changers on the 150 MVA substation transformer and all distribution transformers – AUTO (AVC) with dead-band and time delays, or MANUAL raise / lower
- 15 MVAr capacitor bank for reactive compensation
- Load demand and power-factor set-points, ±1 / ±5 MW load steps, 24-hour daily load profile
- Load models with voltage exponents and frequency sensitivity; motor load in the industrial park
- Energy served and energy-not-served accounting

## Power flow & analysis engine

| | |
|---|---|
| Topology | Node-breaker model with breakers and earthing switches; automatic bus and island detection |
| Load flow | Full Newton-Raphson AC load flow for every electrical island, distributed slack by unit inertia, recalculated every integration step |
| Dynamics | Swing equation per island, turbine/penstock, governor and excitation dynamics; time step 20 ms (5 ms during faults and switching) |
| Fault analysis | Symmetrical components (positive, negative, zero sequence) incl. transformer vector groups; 3PH, L-G, L-L, L-L-G faults with fault resistance at any bus or any % of a line |

## Protection relays

The protection system contains **65 numerical relay functions** (IEC 60255 / ANSI C37.2) evaluated in real time from simulated CT and VT signals. Relays start, time and trip breakers exactly like real devices, including breaker-failure back-up, lock-out (86) and auto-reclose. Each fault automatically creates a **disturbance record** with three-phase current and voltage waveforms and the trip sequence.

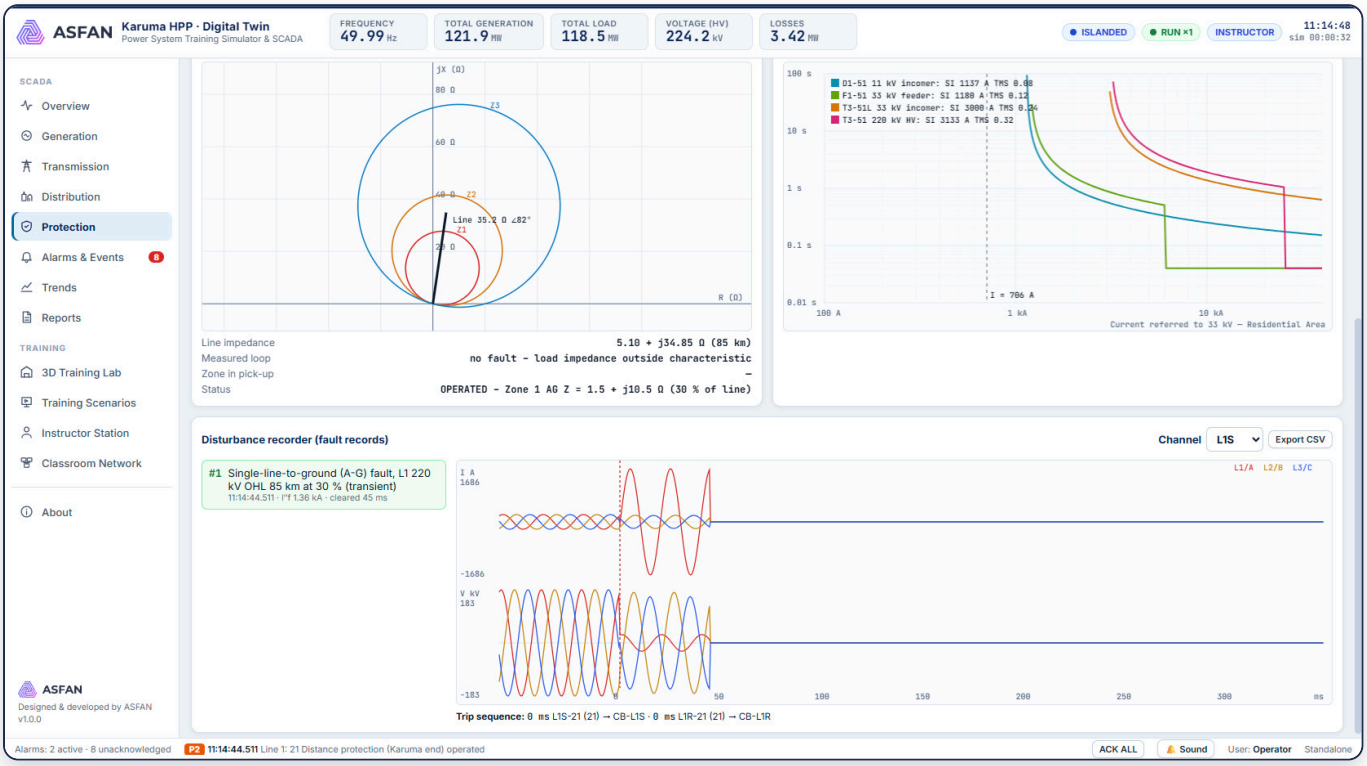



*Figure 5 – Distance-protection R–X diagram, time–current coordination chart and disturbance recorder after a single-phase line fault cleared in 45 ms.*

| Protected zone | Functions (ANSI) |
|---|---|
| Generators | **87G** differential · **64G** stator earth fault · **51V** voltage-restrained O/C · **46** negative sequence · **32** reverse power · **40** loss of excitation · **81U/81O** · **59/27** |
| Transformers | **87T** differential · **50/51** phase O/C (IEC SI/VI/EI curves) · **51N** earth fault · 33 kV and 11 kV incomer O/C |
| Busbars | **87B** busbar differential (Karuma 220 kV, substation 220 kV, 33 kV) |
| 220 kV lines | **21** distance (mho, Z1 80 %, Z2 120 %, Z3 220 %, POTT, six loops with k0) · **67N** directional E/F · **79** auto-reclose with lock-out |
| Grid tie | **67** directional O/C · **81R/78** islanding (ROCOF) · **25** synchro-check |
| Distribution | Feeder **50/51**, **50N/51N** · DT **87T** · 3-stage under-frequency load shedding **81U** · motor **27** · capacitor **59C** |
| All breakers | **50BF** breaker-failure back-up tripping · **86** lock-out with reset · switching interlocks · earthing-switch interlocks |

# SCADA – Alarms, trends & reports

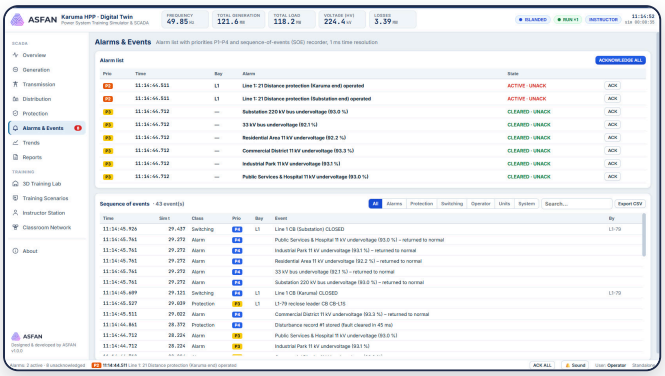


*Figure 6 – Alarm list (P1–P4) and sequence-of-events recorder.*

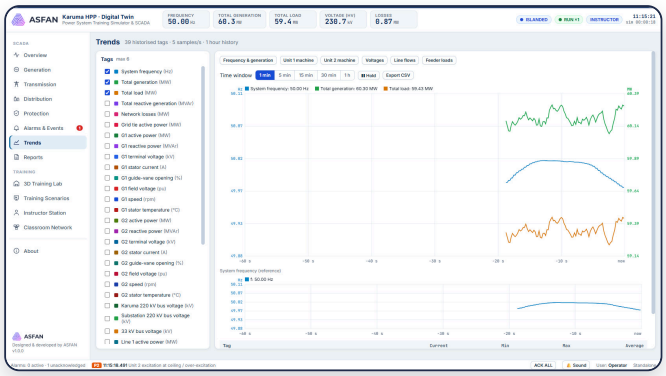


*Figure 7 – Trends: 39 historised tags, up to 6 pens, 1 min – 1 h windows.*

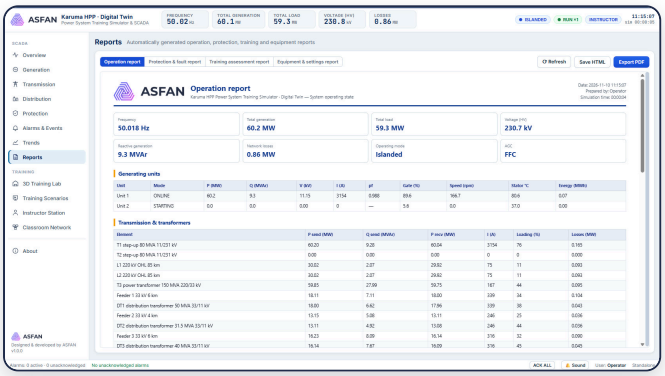


*Figure 8 – Automatic reports with PDF export.*

## Alarms & events

- Four alarm priorities, audible alarm, acknowledge single / all, alarm ticker
- Sequence of events with 1 ms resolution: switching, protection, operator, unit and system events; filter, search and CSV export

## Trends

- 1 hour history at 5 samples per second, hold / resume, statistics (min / max / average), CSV export

## Reports

- Operation report · Protection & fault report · Training assessment report · Equipment & settings report – exported as PDF or HTML

## Training lab, scenarios & instructor station




*Figure 9 – Trainee workstations in the 3D lab: every monitor shows a different live page (generation, transmission, substation, feeders, protection, AGC); click a monitor to zoom in.*

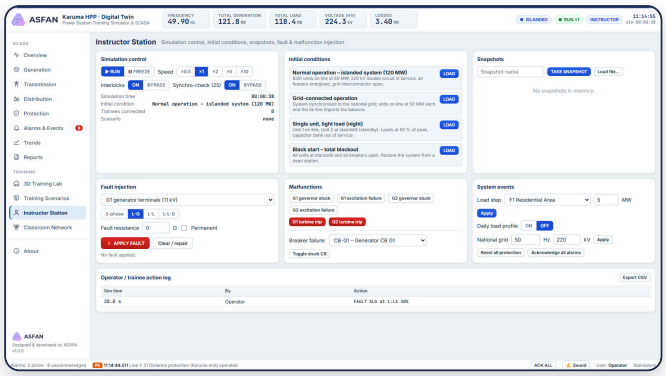



*Figure 10 – Instructor station: run / freeze / speed, initial conditions, snapshots, faults and malfunctions.*

### Training scenarios with automatic assessment

| Scenario | Level | Duration |
|---|---|---|
| Unit start-up & manual synchronisation | BASIC | 12 min |
| Frequency control after a load increase | BASIC | 8 min |
| 220 kV line fault – isolation & restoration | INTERMEDIATE | 15 min |
| Voltage regulation with OLTC, capacitor bank & AVR | INTERMEDIATE | 10 min |
| Synchronisation to the national grid & tie-line control | INTERMEDIATE | 10 min |
| Generator trip, under-frequency load shedding & recovery | ADVANCED | 15 min |
| Black start & system restoration | ADVANCED | 25 min |
| Feeder fault with breaker failure (50BF) | ADVANCED | 12 min |

### Instructor station

- 4 initial conditions: islanded, grid-connected, single unit / light load, total blackout
- Snapshots save / restore (memory and file)
- Fault injection: 3PH, L-G, L-L, L-L-G, fault resistance, transient or permanent, any bus or line position
- Malfunctions: governor stuck, excitation failure, turbine trip, stuck breaker; interlock and synchro-check bypass
- Operator / trainee action log with CSV export

### Classroom network

- Instructor workstation hosts the simulation; trainee workstations join over the training-room LAN
- All trainees see the same live plant and operate it from their own SCADA screens
- Instructor-only commands are protected; each trainee action is logged by name
- Scores, objectives, penalties and quiz answers stored in the training assessment report

## Technical specifications

| Software | |
|---|---|
| Product | Karuma HPP Digital Twin – Power System Training Simulator & SCADA, version 1.0.0 |
| Delivery | Windows installer (Setup .exe) and portable .exe – no additional runtime required |
| Operating modes | Standalone · Instructor workstation (host) · Trainee workstation (client) |
| Simulation | Real time with ×0.25 – ×10 speed and freeze; 20 ms step (5 ms during transients) |
| Data | 39 historised tags (1 h, 5 Hz), 5 000-event SOE buffer, 20 disturbance records |
| Exports | PDF and HTML reports; CSV for events, trends, studies, disturbance records and action log; JSON snapshots |
| Network | TCP/IP LAN, HTTP + Server-Sent Events, default port 47820 |
| Language / units | English user interface, SI units, IEC / ANSI device numbering |

| Recommended workstation | |
|---|---|
| Operating system | Windows 10 or Windows 11, 64-bit |
| Processor / memory | 4-core CPU, 2.5 GHz or faster · 8 GB RAM (16 GB recommended for the instructor workstation) |
| Graphics | DirectX 11 / WebGL 2 capable GPU for the 3D training lab |
| Display | 1920 × 1080 or higher (dual monitors recommended for the instructor) |
| Disk | 500 MB free space |

| Scope of supply | |
|---|---|
| Software | Karuma HPP Digital Twin installer and portable executable |
| Documentation | This product datasheet (included in the installation) |
| Services (optional) | Installation and commissioning, instructor training, additional customer-specific scenarios, model adaptation to site data |

ASFAN

**Software developer: ASFAN**

The Karuma HPP Digital Twin software — simulation models, SCADA, protection, 3D training lab and instructor tools — is **designed and developed by ASFAN**.

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